

Practice Assignment 6

1. The DiscountApplier class (under discount.zip folder) is responsible for applying the new discounts on the prices of the product. If the category of the product is HOME, the new price should now be 90% of the old price. If the category of the product is BUSINESS, the new price should be 110% of the old price.

You will have to use mocks to simulate the ProductDao class, the one responsible for getting all the products from the database.

There is a bug in the implementation. Write the description of the bug inside the file, where you have found that. Submit your test code along with all main codes where you find the bug. **30 Points.**

2. The CustomerPremium class (under CustomerPremium.zip folder) is responsible for applying 20% discounts to the customers, whose previous bill is more than 3000.00 vnd and ordered 5 times or more than 5 times. These customers also considered as Premium customer.

You will have to use mocks to simulate the CustomerDao class, the one responsible for getting all the information of customers from the database.

There is a bug in the implementation. Write the description of the bug inside the file, where you have found that. Submit your test code along with all main codes where you find the bug. **30 Points**

3. Imagine a situation that you have used a new radio/mp3 player. The specification includes: 40 Points

(i) When listening to the radio, pressing the “next” button searches for the next radio station. Pressing “rewind” button searches for the previous radio station. The “mp3” button starts the first mp3-song in memory [when the audio player is just switched on]. Otherwise, it starts the last MP3 song.

(ii) When listening to an mp3-track, pressing “next” button skips to the next song. Pressing “rewind” button plays the previous song. Pressing “radio” button returns to playing the last radio station.

(iii) When listening to the radio or an mp3-song, a traffic alert may fire. The player switches to the radio station broadcasting traffic info. If the traffic is done the player returns to its previous mode (mp3 or radio).

Following figure is the State Diagram of the above situation.

a) Create test cases from the above specification. Execute those test cases on the source codes inside package carAudioPlayer.

b) If you find out any bug, write the bug description in the source file.

c) Submit the source code along with bug description (if any), and the test code.

